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CSIT305 Emerging Information Technologies and their Applications



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Subject Revision



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Subject Overview



Introduction to Emerging IT

- Examples of disruptive technologies
 - Big data
 - Artificial Intelligence and Machine Learning
 - Cloud Services
 - Blockchain
 - 5G Technology
 - Cybersecurity
 - Quantum Computing

Advanced Database Overview

- Introduction to Database Systems
 - Structured/unstructured data, high-quality/low quality data, big data, relational database, non-relational database
- Types of Databases
 - Hierarchical database, network database, NoSQL database, graph database, document database
- Streaming Processing Systems
- Cloud-based Database Systems

Spatial Data and Computing

- Spatial Data
 - Spatial Query Types
 - Window query, range query, k nearest neighbour query
 - Spatial Index Data
 - Quad-tree, R-tree
 - Spatial Query Processing
 - Best first search
- Spatial Textual Data
 - Spatial Textual Index Structure
 - IR-tree
 - Spatial Textual Query Processing
 - Best first search with spatial textual score
- Applications

Graph Data and Computing

- Graph Structure
 - Node, edge, graph
- Graph Representation
 - Undirected vs. directed, unweighted vs. weighted, self-edges (self-loops) graphs vs. multigraph, complete graph, bipartite graph,
 - Adjacency matrix, node degrees, edge list, adjacency list
- Network Properties
 - Degree distribution, path length, distance in a graph, network diameter
- Application: Influence Maximisation in Networks
 - Linear Threshold Model
 - Independent Cascade Model

Text Data and Processing

- Text Data and Mining
- Text Classification vs. Text Clustering
 - Definition, difference, methods
- Natural Language Processing
 - Application scenarios and challenges
 - Core Technologies
 - Language modelling, Part-of-speech tagging, Syntactic parsing, Named-entity recognition, Word sense disambiguation, Semantic role labeling

AI4SE

- Software Project Scheduling & Software Effort Estimation
 - Definition, goal, AI methods
- Requirement Engineering
 - What's the right system to be built?
 - Requirement elicitation
- Code Completion with Software Mining
 - Program analysis key issues, API code recommendation

SE in Cloud Computing I

- Cloud computing fundamentals
 - Definition, properties, service models, deployment models
- Cloud applications
 - Monolithic and microservices
- Requirement engineering for cloud applications
 - Cloud application qualities
 - Key drivers of cloud applications
 - Cloud application requirements

SE in Cloud Computing II

- Monolithic cloud application challenges
 - Technology stack, scalability, change management, deployment
- Microservices: a cloud-based alternative
 - Definition
 - Design principles
 - Microservice deployment
- Define the service architecture

IoT and Edge Computing

- Internet of Things
 - Definition
- Major functional components of an IoT system
 - Edge components
 - Communication layer
 - Cloud
- Design process of an IoT system
- Edge computing
- IoT and edge computing applications

Digital Twin Technology

- What is a Digital Twin?
- Digital Twin Model
 - Data collection
 - Model training
 - Software application
- Digital Twin Type
 - DTP, DTI and DTA
- Digital Twin applications

Blockchain and Smart Contracts

- What is Blockchain?
 - Challenges to be addressed
 - Blockchain basics
 - Major categories of Blockchain uses
- Smart contracts
 - Why smart contracts?
 - DTP, DTI and DTA
 - How to write a smart contract
- Smart contract applications

Emerging Cyber-Attacks

- Recent trends in advanced cyber attacks
 - Types, initial access infection techniques, global distribution, and industry types
- Ransomware
- Botnet attacks using IoT devices
- Cryptojacking
- Cyber attack defense systems

Final Exam



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Basic Info

- Final Exam is worth **50%** of the total marks for the subject.
- To be eligible for a Pass in this subject a student must achieve a mark of at least **40%** in the Final Examination (that is 20 marks out of 50 marks).
- Students who fail to achieve this minimum mark & would have otherwise passed may be given a TF (Technical Fail) for this subject.
- You should review **all topics** we have studied. Any of the topics we have studied in the subject may appear in the exam.



Basic Info

- Final Exam is **3 hours**
- This exam is **CLOSED BOOK - no reference materials permitted.**



Exam Structure

- 8 essay questions in total
 - 6 questions each worth 5 marks, which require short answers
 - 2 questions each worth 10 marks, which require long answers
- Hint
 - Manage your time well
 - Test your understanding, rather than purely memorising the knowledge
 - Focus on quality, rather than quantity



Sample Questions

- Both Quadtrees and R-trees may experience performance degradation under certain data distributions. Explain the underlying reasons for this issue and propose appropriate mitigation strategies for each case.



Sample Questions

- You work for a transportation company that manages a large fleet of vehicles and needs to track routes, schedules, and maintenance histories. Explain how a multi modal database could be used to store and manage this data, and how it would improve performance and efficiency compared to a traditional relational database. Your answer should reference at least two types of data, at least three queries related to the data, and the relevant processing techniques for each query.



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Thank you and Good Luck!

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